

Boss Challenge — Paper 33 (Class 7)

Motion & Time | Electric Current & its Effects | Simple Equations | Fractions & Decimals

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The distance covered by a moving body in one unit of time is called its:
(A) mass
(B) speed
(C) volume
(D) weight
2. A single device that pushes out electricity using the chemicals stored inside it is a:
(A) bulb
(B) switch
(C) cell
(D) wire
3. In maths, using an equals sign to state that two expressions have the same value gives an:
(A) inequality
(B) equation
(C) expression
(D) product
4. The sum of the two halves, $\frac{1}{2} + \frac{1}{2}$, equals:
(A) $\frac{1}{4}$
(B) 1
(C) $\frac{2}{4}$
(D) $\frac{1}{2}$
5. Among these, the most common SI unit used to measure speed is:
(A) metre (m)
(B) kilogram (kg)
(C) metre per second (m/s)
(D) second (s)
6. When two or more cells are joined together to give more push, the combination is called a:
(A) battery
(B) a single cell
(C) a fuse
(D) a filament
7. The value of the variable that makes an equation true is called its:
(A) coefficient
(B) constant
(C) solution (root)
(D) term

8. The product $\frac{1}{2} \times \frac{1}{2}$ equals:

- (A) 1
- (B) $\frac{1}{4}$
- (C) $\frac{1}{2}$
- (D) $\frac{2}{4}$

9. A body that covers equal distances in equal intervals of time is said to be in:

- (A) uniform motion
- (B) rest
- (C) circular motion
- (D) non-uniform motion

10. The thin coiled wire inside a torch bulb that glows brightly when current flows is the:

- (A) terminal
- (B) switch
- (C) filament
- (D) electromagnet

11. Solving the equation $x + 5 = 12$ gives x equal to:

- (A) 17
- (B) 5
- (C) 7
- (D) 60

12. Written as a fraction in simplest form, the decimal 0.5 is:

- (A) $\frac{1}{5}$
- (B) $\frac{5}{1}$
- (C) $\frac{1}{50}$
- (D) $\frac{1}{2}$

13. The instrument fitted in a vehicle to measure the total distance it has travelled is the:

- (A) barometer
- (B) thermometer
- (C) speedometer
- (D) odometer

14. When an electric current passes through a wire and makes it hot, this is called the current's:

- (A) cooling effect
- (B) heating effect
- (C) magnetic effect
- (D) chemical effect

15. Solving the equation $3x = 15$ gives x equal to:

- (A) 45
- (B) 12
- (C) 18
- (D) 5

16. Written as a decimal, the fraction $\frac{1}{4}$ is:

- (A) 0.4
- (B) 4.0
- (C) 0.25
- (D) 0.14

17. The instrument that shows a driver the speed of the vehicle at that moment is the:

- (A) voltmeter
- (B) compass
- (C) speedometer
- (D) odometer

18. A safety device that melts and breaks the circuit when too large a current flows is a:

- (A) switch
- (B) cell
- (C) bulb
- (D) fuse

19. Solving the equation $x - 4 = 10$ gives x equal to:

- (A) 6
- (B) 40
- (C) 10
- (D) 14

20. The sum $\frac{1}{4} + \frac{1}{4}$ equals, in simplest form:

- (A) $\frac{1}{2}$
- (B) $\frac{1}{4}$
- (C) $\frac{1}{8}$
- (D) $\frac{2}{8}$

21. One complete to-and-fro swing of a pendulum, from a point back to the same point, is called one:

- (A) revolution
- (B) rotation
- (C) vibration of sound
- (D) oscillation

22. A coil of wire wound on an iron piece that becomes a magnet only when current flows is an:

- (A) fuse
- (B) permanent magnet
- (C) electromagnet
- (D) filament

23. Solving the equation $x / 2 = 6$ gives x equal to:

- (A) 12
- (B) 8
- (C) 4
- (D) 3

24. The product 0.2×10 equals:

- (A) 0.2
- (B) 0.02
- (C) 20
- (D) 2

25. For a swinging pendulum, the number of seconds needed for one full oscillation is known as its:

- (A) time period
- (B) frequency
- (C) speed
- (D) amplitude

26. An electric bulb in a torch will glow only when the circuit is:

- (A) painted black
- (B) open (broken)
- (C) closed (complete)
- (D) made of glass

27. When a term is moved from one side of an equation to the other, its sign:

- (A) always becomes zero
- (B) changes (+ becomes - and - becomes +)
- (C) doubles in value
- (D) stays exactly the same

28. When you change the fraction $3/5$ into decimal form, the value you get is:

- (A) 0.6
- (B) 5.3
- (C) 0.3
- (D) 0.35

29. The basic SI unit in which we measure time is the:

- (A) kilogram
- (B) second
- (C) hour
- (D) metre

30. Materials such as copper and other metals, which let an electric current pass through them, are called:
- (A) magnets
 - (B) fuses
 - (C) conductors
 - (D) insulators
31. Solving the equation $2x + 3 = 11$ gives x equal to:
- (A) 4
 - (B) 2
 - (C) 14
 - (D) 7
32. One half of 10 is:
- (A) 2
 - (B) 10
 - (C) 20
 - (D) 5
33. To find the SPEED of a body when its distance and time are known, you use:
- (A) speed = distance / time
 - (B) speed = distance x time
 - (C) speed = time / distance
 - (D) speed = distance + time
34. Materials such as rubber and plastic, which do NOT let an electric current pass, are called:
- (A) batteries
 - (B) conductors
 - (C) insulators
 - (D) filaments
35. Adding the same number to BOTH sides of a true equation keeps the equation:
- (A) still true (balanced)
 - (B) impossible to solve
 - (C) equal to zero
 - (D) always false
36. Written as a fraction in simplest form, the decimal 0.75 is:
- (A) $\frac{3}{4}$
 - (B) $\frac{3}{40}$
 - (C) $\frac{7}{5}$
 - (D) $\frac{1}{4}$

37. A car covers 60 kilometres in 2 hours. Its speed is:

- (A) 30 km/h
- (B) 120 km/h
- (C) 2 km/h
- (D) 62 km/h

38. Compared with an old filament bulb giving the same light, a CFL or LED lamp uses:

- (A) exactly the same electricity
- (B) no electricity at all
- (C) more electricity
- (D) less electricity

39. The sentence 'a number increased by 5 gives 12' is written as the equation:

- (A) $x + 5 = 12$
- (B) $x + 12 = 5$
- (C) $5x = 12$
- (D) $x - 5 = 12$

40. Adding the two decimals 2.5 and 1.5 together gives a total of:

- (A) 3.10
- (B) 1.0
- (C) 4.10
- (D) 4.0

41. On a distance-time graph, the line for a body that stays at REST is:

- (A) a vertical straight line
- (B) a wavy curved line
- (C) a horizontal straight line
- (D) a slanting straight line

42. A device used in homes that, like a fuse, breaks the circuit on too much current but can be reset is an:

- (A) a torch bulb
- (B) MCB (miniature circuit breaker)
- (C) a permanent magnet
- (D) a dry cell

43. The sentence 'twice a number is 18' gives the equation $2x = 18$, so the number is:

- (A) 16
- (B) 20
- (C) 36
- (D) 9

44. One third of 9 is:

- (A) 12
- (B) 6
- (C) 3
- (D) 27

45. On a distance-time graph, the line for a body in UNIFORM motion is:

- (A) a circle
- (B) a horizontal straight line
- (C) a zig-zag broken line
- (D) a slanting straight line

46. An electric bell rings because the current flowing through its coil turns the coil into an:

- (A) a cell
- (B) electromagnet
- (C) insulator
- (D) a fuse

47. To CHECK that $x = 3$ solves the equation $x + 4 = 7$, you should:

- (A) multiply both sides by 3
- (B) put 3 in place of x and see if both sides are equal
- (C) change the 7 to a 3
- (D) rub out the $+ 4$ sign

48. Written as a fraction in simplest form, the decimal 0.1 is:

- (A) $\frac{1}{1}$
- (B) $\frac{10}{1}$
- (C) $\frac{1}{100}$
- (D) $\frac{1}{10}$

49. A train covers 150 kilometres in 3 hours at a steady speed. Its speed is:

- (A) 3 km/h
- (B) 50 km/h
- (C) 153 km/h
- (D) 450 km/h

50. Bringing a compass needle near a wire carrying current makes the needle move. This shows the current's:

- (A) chemical effect
- (B) lighting effect
- (C) magnetic effect
- (D) heating effect

51. Solving the equation $4x - 1 = 11$ gives x equal to:
- (A) 3
 - (B) 2.5
 - (C) 12
 - (D) 48
52. The product 0.3×0.2 equals:
- (A) 0.06
 - (B) 6
 - (C) 0.6
 - (D) 0.5
53. A speed of 10 m/s tells us that the body, each second, covers:
- (A) 100 metres
 - (B) 10 seconds
 - (C) 10 kilometres
 - (D) 10 metres
54. Plugging too many appliances into one socket, drawing a very large current, is called:
- (A) earthing
 - (B) insulating
 - (C) overloading
 - (D) charging
55. A car's steady speed s satisfies the equation $2s = 60$ (km/h). The speed s is:
- (A) 120 km/h
 - (B) 62 km/h
 - (C) 30 km/h
 - (D) 58 km/h
56. A body covers 3.6 metres in 2 seconds at a steady speed. Its speed is:
- (A) 5.6 m/s
 - (B) 2 m/s
 - (C) 7.2 m/s
 - (D) 1.8 m/s
57. To find the TIME taken when the distance and the speed are known, you use:
- (A) time = distance / speed
 - (B) time = distance $\times$ speed
 - (C) time = speed / distance
 - (D) time = speed $\times$ distance
58. The heating element of an electric room heater is made of a special wire because that wire:
- (A) stays completely cold
 - (B) becomes very hot without melting
 - (C) turns into a magnet
 - (D) blocks all current

59. Solving the equation $x + 7 = 7$ gives x equal to:

- (A) 14
- (B) 1
- (C) 0
- (D) 7

60. The reciprocal of the fraction $\frac{2}{3}$ is:

- (A) $-\frac{2}{3}$
- (B) $\frac{2}{3}$
- (C) $\frac{3}{2}$
- (D) 6

61. To find the DISTANCE covered when the speed and the time are known, you use:

- (A) distance = speed + time
- (B) distance = speed / time
- (C) distance = time / speed
- (D) distance = speed x time

62. When the current to an electromagnet is switched OFF, the electromagnet:

- (A) loses its magnetism
- (B) starts to give light
- (C) becomes a stronger magnet
- (D) turns into a permanent magnet

63. Solving the equation $10 - x = 4$ gives x equal to:

- (A) 6
- (B) 4
- (C) 14
- (D) 40

64. Written as a decimal, the fraction $\frac{7}{10}$ is:

- (A) 10.7
- (B) 0.07
- (C) 0.7
- (D) 7.10

65. A pendulum makes 20 complete oscillations in 40 seconds. Its time period is:

- (A) 2 seconds
- (B) 40 seconds
- (C) 20 seconds
- (D) 800 seconds

66. Compared with one cell, two cells joined the correct way in a torch usually make the bulb glow:

- (A) dimmer
- (B) a different colour
- (C) brighter
- (D) not at all

67. The process of moving a term to the other side of an equation, changing its sign, is called:

- (A) transposition (transposing)
- (B) rounding off
- (C) multiplication
- (D) factorising

68. The product 0.25×4 equals:

- (A) 8
- (B) 0.100
- (C) 0.29
- (D) 1

69. Compared with an old sand-clock or sundial, a modern quartz wrist-watch is:

- (A) much less accurate
- (B) exactly as accurate
- (C) useless for telling time
- (D) much more accurate

70. The wire inside a fuse is chosen to have a LOW melting point so that it:

- (A) melts quickly when the current is too high
- (B) never melts at all
- (C) conducts no current
- (D) glows to give light

71. Solving the equation $x / 5 = 2$ gives x equal to:

- (A) 3
- (B) 10
- (C) 7
- (D) 0.4

72. The sum $1/5 + 2/5$ equals:

- (A) $3/25$
- (B) $3/10$
- (C) $2/5$
- (D) $3/5$

73. A body whose speed keeps changing from moment to moment is moving in:

- (A) straight-line uniform motion
- (B) non-uniform motion
- (C) rest
- (D) uniform motion

74. In a circuit diagram, the device drawn as a small circle with a loop or cross inside stands for a:

- (A) cell
- (B) bulb
- (C) wire
- (D) switch

75. Solving the equation $2x - 6 = 0$ gives x equal to:

- (A) 6
- (B) 12
- (C) 3
- (D) 0

76. Of the two decimals 0.5 and 0.45, the LARGER one is:

- (A) they are equal
- (B) 0.5
- (C) neither is a real number
- (D) 0.45

77. An ancient device that tells the time of day from the shadow cast by the Sun is a:

- (A) sundial
- (B) speedometer
- (C) pendulum clock
- (D) quartz watch

78. For safety, you should never touch an electric switch or appliance when your hands are:

- (A) warm from the Sun
- (B) wet
- (C) wearing rubber gloves
- (D) clean and dry

79. A bulb circuit uses n cells, and one more cell would make 3 cells in all: $n + 1 = 3$. The value of n is:

- (A) 1
- (B) 3
- (C) 2
- (D) 4

80. Written in simplest form, the fraction $\frac{2}{4}$ is:

- (A) $\frac{2}{4}$
- (B) $\frac{1}{2}$
- (C) $\frac{4}{2}$
- (D) $\frac{1}{4}$

81. A body moves at a steady 5 m/s for 12 seconds. The distance it covers is:

- (A) 60 metres
- (B) 2.4 metres
- (C) 5 metres
- (D) 17 metres

82. In the symbol for a single cell, the longer thin line stands for the:
- (A) switch
 - (B) negative terminal
 - (C) fuse
 - (D) positive terminal
83. The sentence 'a number divided by 3 gives 4' becomes $x / 3 = 4$, so the number is:
- (A) 1
 - (B) 12
 - (C) 7
 - (D) 4
84. The product 1.2×3 equals:
- (A) 3.6
 - (B) 4.2
 - (C) 3.2
 - (D) 36
85. The faster of two bodies, given the same length of time, will always cover:
- (A) no distance at all
 - (B) less distance
 - (C) exactly the same distance
 - (D) more distance
86. The main job of a switch in an electric circuit is to:
- (A) turn the wire into a magnet
 - (B) open or close the circuit to stop or start the current
 - (C) store electricity for later
 - (D) produce the current itself
87. A pendulum's time period t satisfies $3t = 6$ (seconds). The time period t is:
- (A) 3 seconds
 - (B) 18 seconds
 - (C) 9 seconds
 - (D) 2 seconds
88. One half of 0.6 is:
- (A) 0.3
 - (B) 0.06
 - (C) 1.2
 - (D) 0.12
89. A simple pendulum is useful for measuring time mainly because its time period stays:
- (A) completely random each swing
 - (B) longer with every swing
 - (C) almost the same for each swing
 - (D) shorter with every swing

90. A torch bulb is connected to a cell with a switch and wires, but it will not light. A likely simple cause is that:

- (A) the bulb is glowing too brightly
- (B) the circuit is broken somewhere (e.g. switch open or loose wire)
- (C) there is too much daylight in the room
- (D) the wires are the wrong colour

91. Solving the equation $7 + 2x = 13$ gives x equal to:

- (A) 3
- (B) 10
- (C) 20
- (D) 6

92. Written as a decimal, the fraction $5/100$ is:

- (A) 0.005
- (B) 0.05
- (C) 0.5
- (D) 5.0

93. The unit 'km/h', often seen on road signs, is used to measure the:

- (A) mass of vehicles
- (B) speed of vehicles
- (C) time of a journey
- (D) fuel in the tank

94. The heating effect of current is put to good use in an appliance such as an:

- (A) compass
- (B) electric bell
- (C) loudspeaker
- (D) electric iron

95. Solving the equation $5x = 5$ gives x equal to:

- (A) 25
- (B) 0
- (C) 1
- (D) 5

96. The sum $1/2 + 1/4$ equals, in simplest form:

- (A) $1/6$
- (B) $2/4$
- (C) $3/4$
- (D) $2/6$

97. The shape of a distance-time graph is useful because, at a glance, it tells us whether the motion is:

- (A) the price of the fuel
- (B) the driver's name
- (C) the colour of the vehicle
- (D) uniform or non-uniform

98. In a series circuit with a bulb, if a fuse rated at 5 amperes carries 0.5 of that rating, the current flowing is:

- (A) 5 amperes
- (B) 0.5 amperes
- (C) 10 amperes
- (D) 2.5 amperes

99. Solving the equation $3(x) = 0$, where $3(x)$ means 3 times x , gives x equal to:

- (A) 9
- (B) 0
- (C) 1
- (D) 3

100. A fuse is rated at 6 amperes and the circuit draws $\frac{1}{3}$ of that rating. The current flowing is:

- (A) 3 amperes
- (B) 2 amperes
- (C) 18 amperes
- (D) 6 amperes